TEXORA · COURSE NOTES · LECTURE SHEET

Introduction to Textile

What a textile is, the building blocks it is made of, how the craft developed over thousands of years, and how raw fibre becomes a finished product.

TE-1109 · FUNDAMENTALS OF TEXTILE ENGINEERING · TOPIC 1 · LEVEL 1 TERM 1

1. What is a textile?

The word *textile* comes from the Latin verb *texere*, meaning “to weave”, and from the Latin adjective *textilis*, meaning “woven”. In its oldest and narrowest sense, a textile is a cloth produced by weaving — one set of yarns (the warp) interlaced at right angles with another set (the weft) on a loom. That definition was adequate for centuries, because weaving was the dominant way of converting fibre into a usable sheet material.

It is no longer adequate. Knitting builds fabric by interlooping yarns; braiding interlaces them diagonally; felting and bonding make fabric directly from fibres without any yarn at all. None of these is weaving, yet all of their products are textiles. The modern definition is therefore wider:

> **DEFINITION**
>
> A textile is a flexible material made from fibres or yarns — by weaving, knitting, braiding, felting, bonding or similar processes — and used for clothing, household goods, and a wide range of technical and industrial purposes.

Three ideas inside this definition deserve attention. First, the starting material is the **fibre**: a thin, hair-like unit whose length is very large compared with its diameter. Second, fibres are usually — but not always — converted into **yarn** before they become fabric; nonwoven fabrics are the important exception. Third, the product is **flexible**. Drape, softness and the ability to fold without breaking separate textiles from rigid sheet materials such as plastic film or metal foil, although some modern technical textiles are engineered to be extremely strong.

Clothing is also one of the three basic needs of human life, alongside food and shelter. It protects the body from heat, cold, sun, wind and injury. With time it took on further roles — comfort, modesty, identity, status and fashion — but protection came first, and it is still the foundation on which the entire industry rests.

1.1 Characteristics of textile materials

Whatever the end use, textile materials share a recognisable set of characteristics:

- **High length-to-diameter ratio.** The fibre, the basic unit, is at least a hundred times longer than it is thick. This fineness is what makes flexible structures possible.
- **Flexibility and drape.** Textiles bend, fold and conform to the shape of the body or object they cover.
- **Adequate strength.** A textile must survive the tensions of manufacturing and of use — pulling, rubbing, washing and repeated bending.
- **Porosity and comfort.** The spaces between fibres and yarns let air and moisture vapour pass, which is why fabrics feel comfortable against the skin.
- **Workability.** Textiles can be cut, sewn, joined and shaped with simple tools, so they can be turned into an almost unlimited range of products.

2. The three building blocks: fibre, yarn and fabric

Nearly everything in textile engineering is a structure built from fibres, yarns or fabrics. It is worth fixing the three terms precisely at the very beginning, because the rest of the degree uses them constantly.

UNIT	WHAT IT IS	EXAMPLES
Fibre	The smallest unit of a textile material: a thin, flexible, hair-like strand with a length many times its diameter. Fibres may be short (*staple* fibres) or continuous (*filaments*).	Cotton, jute, wool, silk; polyester, nylon, viscose
Yarn	A continuous strand made by twisting fibres together (spun yarn) or by assembling filaments (filament yarn). Yarn is strong enough to be woven or knitted.	Ring-spun cotton yarn, rotor yarn, polyester filament yarn
Fabric	A sheet-like material made from yarns (by weaving, knitting or braiding) or directly from fibres (by felting or bonding). Fabric is what is cut and sewn into products.	Woven poplin, knitted single jersey, nonwoven interlining

The relationship is a hierarchy: fibres are assembled into yarns, and yarns are assembled into fabrics. Two measurements recur throughout the degree and are introduced here. **Yarn count** expresses how fine or coarse a yarn is; **GSM** (grams per square metre) expresses how heavy a fabric is. Both are studied in detail in later courses.

3. A short chronological development

Textile making is older than writing, older than the wheel, older than metalworking. The main stages of its long development are summarised in the table below, and the most important turning points are discussed after it.

PERIOD	DEVELOPMENT
Prehistoric era	Early humans cover themselves with animal skins, bark and leaves. With settled farming, plant and animal fibres are spun with hand spindles: flax is cultivated and woven into linen in Egypt and Mesopotamia; cotton is grown and woven in the Indus Valley; silk production begins in China.
Ancient world	Spinning and loom weaving spread across Asia and Europe. Bengal’s handwoven cotton muslin — so fine it was called “woven air” — is traded across the Roman world. Wool becomes the main fibre of Europe.
Middle ages	Textile production is organised by craft guilds. The spinning wheel, arriving in Europe from Asia, multiplies the output of the hand spinner several times over.
1733 – 1785	The Industrial Revolution begins in the textile industry. The flying shuttle (1733) doubles weaving speed; the spinning jenny (1764), water frame (1769) and spinning mule (1779) mechanise spinning; the power loom (1785) mechanises weaving. Production moves from cottages into factories.
19th century	Steam power, ring spinning, the sewing machine and the first synthetic dyes cut costs dramatically. Raw cotton, wool and finished cloth are traded globally.
20th century	Man-made fibres arrive: viscose rayon first, then nylon (1938), polyester and acrylic. High-speed machinery, synthetic dyes and easy-care finishes transform both output and products.
Today	Automation, computer-aided design and manufacturing, and a fast-growing field of technical textiles — fabrics that filter, conduct electricity, protect against fire and bullets, support surgery, or sense and react to their surroundings.

3.1 The inventions that changed everything

Between 1733 and 1785 a handful of inventions broke a bottleneck that had lasted for millennia. Weavers could always use yarn faster than spinners could supply it; each invention attacked one side of that imbalance:

INVENTION	YEAR	WHAT IT CHANGED

Flying shuttle — John Kay	1733	A shuttle carrying the weft is knocked across the loom by a cord instead of being passed by hand. One weaver can weave wide cloth, roughly twice as fast.
Spinning jenny — James Hargreaves	1764	One wheel drives many spindles at once (eventually over a hundred), so one spinner produces many yarns simultaneously.
Water frame — Richard Arkwright	1769	Rollers draw the fibres out and water power twists them: a stronger yarn, spun continuously in a factory.
Spinning mule — Samuel Crompton	1779	Combines the jenny and the water frame to spin yarn that is both fine and strong — suitable for the finest muslins.
Power loom — Edmund Cartwright	1785	Weaving is driven by power instead of by the weaver's hands and feet, completing the move of both spinning and weaving into mills.

Notice the pattern across the whole history. For thousands of years the *methods* barely changed — only skill, organisation and scale grew. Then, within about fifty years, machinery changed everything, and textiles became the first truly industrialised sector of the modern economy. This is why the textile industry is often called the mother of industrialisation, and why so many later industries — engineering, chemicals, logistics — first grew up in its service.

4. Classification of textiles

Textiles can be classified in several ways. Two classifications are used constantly: by **method of production** and by **end use**.

4.1 By method of production

- **Woven fabrics** — two sets of yarns interlaced at right angles (warp and weft) on a loom. Generally firm, stable and strong: shirting, denim, suiting.
- **Knitted fabrics** — yarns formed into loops and interlocked. Stretchy and comfortable: T-shirts, sweaters, hosiery, underwear.
- **Nonwoven fabrics** — a web of fibres bonded together mechanically, thermally or chemically, with no yarn stage. Cheap and versatile: wipes, masks, interlinings, geotextiles.
- **Braided and lace fabrics** — yarns interlaced diagonally or patterned openly: shoelaces, ropes, decorative trims.
- **Tufted and composite fabrics** — yarns punched through a base fabric (carpets) or layers combined for special performance.

4.2 By end use

- **Apparel textiles** — clothing of every kind, from everyday wear to fashion and protective clothing. This is the largest and most familiar group.
- **Household (home) textiles** — bed linen, towels, curtains, upholstery, carpets and table linen.
- **Technical textiles** — fabrics chosen for what they *do* rather than how they look. The field is conventionally divided into twelve application areas:

AREA	TYPICAL USES	AREA	TYPICAL USES
Agrotech	Crop covers, shade nets, fishing nets	Indutech	Filters, conveyor belts, abrasives
Buildtech	Scaffolding nets, architectural membranes	Medtech	Bandages, sutures, implants
Clothtech	Shoe components, interlinings, zips' tapes	Mobiltech	Seat belts, airbags, tyre cord
Geotech	Soil stabilisation, erosion control	Oekotech	Oil-absorbent mats, waste recycling
Hometech	Furniture fabrics, mattress components	Packtech	Sacks, bags, wrapping fabrics
Protech	Fire, bullet and chemical protection	Sportech	Sports shoes, tents, parachutes

A textile engineer may spend a career in any one of these areas. The physics and chemistry are the same; only the performance requirements change.

5. From fibre to finished product: the textile chain

Every textile product, whether a handloom sari or a factory-made shirt, passes through the same basic chain of stages. Understanding this chain is the foundation of the whole subject — almost every later course in this degree studies one link of it in depth.

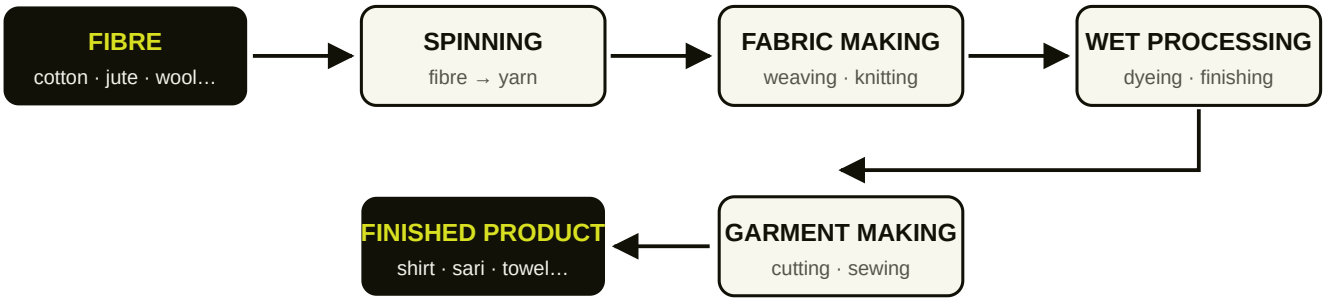


Figure 1 — The basic process flow of textile materials, from raw fibre to finished product.

5.1 Fibre — where the chain begins

The chain begins with raw fibres. Natural fibres come from plants (cotton, jute, flax), animals (wool, silk) or minerals; man-made fibres are manufactured from polymers, either regenerated from natural sources (viscose) or synthesised from petrochemicals (polyester, nylon, acrylic). Fibre properties — length, fineness, strength, absorbency — set the limits of everything that follows. A mill cannot spin a fine, strong yarn from short, weak fibre, whatever machines it owns.

5.2 Spinning — fibre into yarn

Spinning opens and cleans the fibres, aligns them into a parallel strand, draws the strand out to the required fineness (drafting) and twists it (twisting) so that the fibres grip one another and form a continuous yarn with useful strength. The full sequence — blowroom, carding, drawing, roving and the ring frame — is the subject of the spinning technology courses.

5.3 Fabric making — yarn into fabric

Yarns are converted into fabric mainly by weaving or knitting, or the fabric is formed directly from fibre in nonwoven production. At this point the material is called **grey** or **greige** fabric: unfinished, usually off-white, and not yet fit for the consumer.

METHOD	PRINCIPLE	CHARACTER OF THE FABRIC
Weaving	Warp and weft yarns interlaced at right angles	Firm, dimensionally stable, strong in both directions
Knitting	Yarns formed into loops and interlocked row by row	Elastic, soft, comfortable; less stable than woven fabric
Nonwoven	Fibre web bonded by needles, heat or chemicals	Low cost, isotropic, often disposable or hidden in use

5.4 Wet processing — giving fabric its colour and finish

Grey fabric is transformed in three groups of operations. **Preparation** (desizing, scouring and bleaching) removes natural and added impurities so the fabric is clean, white and absorbent. **Colouration** (dyeing and printing) gives it colour and pattern. **Finishing** (softening, mercerising, crease-resist and many other treatments) gives it the handle, appearance and performance the buyer has specified. Wet processing uses the most water, chemicals and energy of any stage, which is why it is also the industry's biggest environmental concern.

5.5 Garment making — fabric into product

The finished fabric is turned into a product through a sequence of operations: a **pattern** is made for each component of the garment; a **marker** arranges the pattern pieces on the fabric with minimum waste; the fabric is **spread** in layers and **cut**; the pieces are **sewn** together, trimmed and provided with buttons, zips and labels; finally the garment is **pressed, inspected and packed**. Household and technical products follow the same logic with different shapes and joining methods.

WORKED EXAMPLE — ONE COTTON POLO SHIRT

Follow a single product through the chain. Cotton fibres about 28 mm long are opened, cleaned and carded, then spun into yarn. The yarn is knitted into a piqué fabric of roughly 200 g/m². The grey fabric is scoured, bleached, dyed to the buyer's shade, softened and compacted. The finished fabric is cut into body, sleeve and collar pieces, sewn into a polo shirt, checked, pressed and packed. Five stages, dozens of machines, and hundreds of individual operations stand behind one ordinary shirt — and a fault at any stage (weak yarn, uneven dyeing, a skipped stitch) can make the whole shirt unsaleable.

KEY POINT

Each stage adds value and cost. A fault introduced at an early stage — poor fibre, weak yarn — cannot be repaired later; it can only be carried, expensively, to the end. This is why quality control in textiles starts at the fibre, not at the final inspection.

6. The textile industry and Bangladesh

Textiles and clothing together form one of the largest manufacturing industries in the world, employing tens of millions of people and supplying a product that every human being uses every day. For Bangladesh the industry is far more than large: ready-made garments (RMG) account for the great majority — more than four-fifths — of the country's export earnings, and Bangladesh is the world's second-largest exporter of apparel after China. Millions of workers, a large share of them women, are employed in several thousand garment factories, supported by spinning, weaving, knitting, dyeing and

finishing mills behind them. Bangladesh is also the world's leading producer and exporter of raw jute, the "golden fibre".

This is the industry a textile engineer in Bangladesh enters: spinning mills, composite knitwear factories, denim mills, dyeing and washing plants, buying houses and testing laboratories — with roles in production, quality, merchandising, research and management. The courses of this degree map directly onto those workplaces, one stage of the chain at a time.

7. Glossary — key terms of this topic

TERM	MEANING
Fibre	The basic unit of a textile: a thin, flexible strand whose length is many times its diameter.
Staple fibre	A fibre of limited, short length (cotton, wool, cut man-made fibres), measured in millimetres or inches.
Filament	A fibre of effectively unlimited length, as produced for silk and man-made yarns.
Yarn	A continuous strand of twisted fibres or assembled filaments, suitable for weaving or knitting.
Warp / weft	In a woven fabric, the lengthwise yarns (warp) and the crosswise yarns (weft) interlaced with them.
Grey (greige) fabric	Fabric as it leaves the loom or knitting machine, before any preparation, dyeing or finishing.
Spinning	The process of converting fibres into yarn by cleaning, drafting and twisting.
Wet processing	All water-based treatments of fabric: preparation, dyeing, printing and finishing.
Yarn count	A number expressing the fineness or coarseness of a yarn.
GSM	Grams per square metre — the standard measure of fabric weight.
Technical textiles	Textiles engineered primarily for functional performance rather than appearance.

8. Key points and summary

- A textile is a flexible material built from fibres or yarns; the word comes from the Latin *texere*, "to weave", but the modern meaning covers knitting, braiding, felting and bonding as well.
- The building blocks form a hierarchy: **fibre → yarn → fabric**. Yarn count measures yarn fineness; GSM measures fabric weight.

- The craft is ancient — linen, cotton and silk are thousands of years old — but its transformation came with the machines of 1733–1785, which made textiles the first industry of the Industrial Revolution.
- Textiles are classified by production method (woven, knitted, nonwoven, braided) and by end use (apparel, household, and technical textiles in twelve application areas).
- Production follows a fixed chain: **fibre → spinning → fabric making → wet processing → garment making → finished product**. Almost every course in textile engineering studies one stage of this chain in depth.
- Quality is built in from the fibre stage; later stages cannot repair early faults. In Bangladesh, this industry — led by RMG exports and jute — is the backbone of the national economy and the workplace of the textile engineer.

Practice

Answer in your own words; brief model answers follow.

1. Define a textile in one sentence that would also cover a fishing net and a seat belt, and name what your definition deliberately includes.
2. Trace the full chain for a cotton towel, naming the material form at every stage from plant to shop.
3. A fabric can be knitted, woven or non-woven. For each construction, name one product where that construction is the right choice and say why.
4. Classify these by end use (apparel, household, technical — with the technical sub-area): a rain jacket, a hospital bandage, a carpet, an airbag, a bed sheet, a geotextile.
5. Why did the power loom and spinning machinery appear in Britain first, and what single invention of 1733 started the chain of mechanisation?
6. Bangladesh imports almost all its raw cotton yet exports garments worldwide. Using the chain, explain why this is workable — which stages sit inside the country and which outside.

Answer key

1. Example: “A textile is a flexible sheet or structure made from fibres or yarns” — it deliberately includes nets and belts because they are yarn/fibre structures, not only clothing.
2. Cotton plant → seed cotton → (ginning) lint bale → (spinning) yarn → (weaving) grey fabric → (dyeing/finishing, terry weave) towel fabric → (cutting/hemming) finished towel.
3. Woven — a shirt (stable, crisp). Knitted — a T-shirt (stretch and comfort). Non-woven — a disposable wipe or mask (cheap, no yarn stage needed).
4. Rain jacket — apparel (or protective technical); bandage — medical technical; carpet — household/interior; airbag — automotive technical; bed sheet — household; geotextile —

civil/geotechnical technical.

5. Coal, capital, colonies supplying cotton and markets, and a machine-making culture converged there; the flying shuttle (John Kay, 1733) began the sequence.
6. Fibre production sits outside; spinning, knitting/weaving, dyeing and garment making sit inside — the country imports the chain’s first input and performs every value-adding stage after it.
